Problem Set 24

Arc Length

November 5, 2026

Problem 1

(a)

$$n = 1 \implies l = \sqrt{2} = 1.414$$
$$n = 2 \implies l = 1.4604$$
$$n = 4 \implies l = 1.474$$

(b) By the triangle inequality,

$$AB \leq AC + BC$$

Hence, every polygon length is an underestimate compared to L.

(c)

$$\Delta y = f'(x)\Delta x$$

(d)

$$y = mx + c \implies y' = m$$
$$L = \int_a^b \sqrt{1 + m^2}\, dx = (b - a)\sqrt{1 + m^2}$$
$$y = \sqrt{1 - x^2} = \boxed{\frac{\pi}{6}}$$

Problem 2

(b)

$$x = y^2 \implies x' = 2y$$
$$l = \int_0^1 \sqrt{1 + 4y^2}\, dy$$
$$\sqrt{1 + 4y^2} = \sec\theta$$
$$y = \frac{1}{2}\tan\theta \implies dy = \frac{1}{2}\sec^2\theta\, d\theta$$
$$l = \frac{1}{2}\int_0^{\arctan 2} \sec^3\theta\, d\theta$$
$$= \frac{1}{4}(\sec\theta\tan\theta + \ln|\sec\theta + \tan\theta|)\Big|_0^{\arctan 2}$$

$$= \frac{1}{4}(\sqrt{5} \cdot 2 + \ln(2+\sqrt{5})$$
$$= \boxed{\frac{\sqrt{5}}{2} + \frac{1}{4}\ln(2+\sqrt{5})}$$

Problem 3

(b)

$$x = y + y^3 \implies x' = 1 + 3y^2$$
$$\boxed{l = \int_1^4 \sqrt{1 + (1+3y^2)^2}\, dy}$$

(c) $\sqrt{9+66^2}$ gives the straight-line distance between the two points, functioning as the lower bound. $3 + 66$ gives the simpler, overestimate version of the integral, functioning as the upper bound.

Problem 4

(b)

$$x = \frac{1}{3}\sqrt{y}(y-3) = \frac{1}{3}y^{\frac{3}{2}} - y^{\frac{1}{2}}$$
$$x' = \frac{1}{2}(y^{\frac{1}{2}} - y^{-\frac{1}{2}})$$
$$l = \int_1^9 \sqrt{1 + \left(\frac{1}{2}(y^{\frac{1}{2}} - y^{-\frac{1}{2}})\right)^2}\, dy$$
$$= \int_1^9 \sqrt{1 + \frac{1}{4}(y - 2 + \frac{1}{y})}\, dy$$
$$= \int_1^9 \frac{1}{2}\sqrt{y + 2 + \frac{1}{y}}\, dy$$
$$= \frac{1}{2}\int_1^9 \sqrt{y} + \frac{1}{\sqrt{y}}\, dy$$
$$= \frac{1}{3}y^{\frac{3}{2}} + y^{\frac{1}{2}}\Big|_1^9$$
$$= 9 + 3 - \frac{1}{3} - 1$$
$$= \boxed{\frac{32}{3}}$$

(c)

$$y = \ln\sec x \implies y' = \sec x \tan x \cdot \frac{1}{\sec x} = \tan x$$
$$l = \int_0^{\frac{\pi}{4}} \sqrt{1 + \tan^2 x}\, dx$$

$$= \int_0^{\frac{\pi}{4}} \sec x \, dx$$
$$= \ln|\sec x + \tan x| \Big|_0^{\frac{\pi}{4}}$$
$$= \boxed{\ln(\sqrt{2}+1)}$$

(e)

$$1 + (A-B)^2 = A^2 - 2AB + b^2 + 4AB = (A+B)^2$$
$$y = \frac{x^2}{8} - \ln x$$
$$y' = \frac{1}{4}x - \frac{1}{x}$$
$$l = \int_1^e \sqrt{1 + \left(\frac{1}{4}x - \frac{1}{x}\right)^2} \, dx$$
$$= \int_1^e \frac{1}{4}x + \frac{1}{x} \, dx$$
$$= \frac{x^2}{8} + \ln x \Big|_1^e$$
$$= \boxed{\frac{e^2+7}{8}}$$

Problem 5

$$x^{\frac{2}{3}} + y^{\frac{2}{3}} = 1 \implies y^{\frac{2}{3}} = 1 - x^{\frac{2}{3}}$$
$$y = \left(1 - x^{\frac{2}{3}}\right)^{\frac{3}{2}}$$
$$y' = \frac{3}{2}(1 - x^{\frac{2}{3}})^{\frac{1}{2}}(-\frac{2}{3}x^{-\frac{1}{3}})$$

$$y' = -\frac{(1 - x^{\frac{2}{3}})^{\frac{1}{2}}}{x^{\frac{1}{3}}}$$
$$y = x \implies x^{\frac{2}{3}} = \frac{1}{2} \implies x = \frac{1}{2\sqrt{2}}$$
$$l = \int_{\frac{1}{2\sqrt{2}}}^1 \sqrt{1 + \frac{1 - x^{\frac{2}{3}}}{x^{\frac{2}{3}}}} \, dx$$
$$= \int_{\frac{1}{2\sqrt{2}}}^1 x^{-\frac{1}{3}} \, dx$$
$$= \frac{3}{2}x^{\frac{2}{3}} \Big|_{\frac{1}{2\sqrt{2}}}^1$$

$$= \frac{3}{2} - \frac{3}{4}$$
$$= \boxed{\frac{3}{4}}$$
$$8 \cdot \frac{3}{4} = \boxed{6}$$

Near $(0, 1)$, the derivative approaches negative infinity, rendering the problem more complicated to calculate.

Problem 6

(a)

$$y = 180 - \frac{1}{45}x^2 \implies y' = -\frac{2}{45}x$$
$$l = \int_0^{90} \sqrt{1 + \frac{4}{2025}x^2}\, dx$$
$$x = \frac{45}{2}\tan\theta \implies dx = \frac{45}{2}\sec^2\theta\, d\theta$$
$$l = \int_0^{\arctan 4} \frac{45}{2}\sec^3\theta\, d\theta$$
$$= \frac{45}{4}(\sec x \tan x + \ln|\sec x + \tan x|)\Big|_0^{\arctan 4}$$
$$= \frac{45}{4}(\sqrt{17} \cdot 4 + \ln(\sqrt{17} + 4))$$
$$= \boxed{45\sqrt{17} + \frac{45}{4}\ln(\sqrt{17} + 4) \approx 209.1\text{m}}$$

(b) The values are the absolute upper and lower bounds to length.

Problem 7

(a)

$$y = \int_1^x \sqrt{t^3 - 1}\, dt \implies y' = \sqrt{x^3 - 1}$$
$$l = \int_1^4 \sqrt{1 + x^3 - 1}\, dx$$
$$= \frac{2}{5}x^{\frac{5}{2}}\Big|_1^4$$
$$= \boxed{\frac{62}{5}}$$

(b)

$$\frac{e^t - e^{-t}}{2} = \int_0^t \sqrt{1 + f'(x)^2}\, dx$$

$$\frac{e^t + e^{-t}}{2} = \sqrt{1 + f'(x)^2}$$
$$f'(x)^2 = \frac{1}{4}(e^{2t} + 2 + e^{-2t} - 1)$$
$$f'(x) = \frac{e^x - e^{-x}}{2}$$
$$f(x) = \int \frac{e^x - e^{-x}}{2}\,dx$$
$$\boxed{f(x) = \frac{e^x + e^{-x}}{2}}$$

Problem 8

(a)

$$y = \frac{2}{3}x^{\frac{3}{2}} \implies y' = \sqrt{x}$$
$$l = \int_0^x \sqrt{1+t}\,dt$$
$$= \int_1^{x+1} \sqrt{u}\,du$$
$$= \frac{2}{3}u^{\frac{3}{2}}\Big|_1^{x+1}$$
$$= \boxed{\frac{2}{3}(x+1)^{\frac{3}{2}} - \frac{2}{3}}$$

(b)

$$\frac{2}{3}(x+1)^{\frac{3}{2}} - \frac{2}{3} = \frac{14}{3}$$
$$(x+1)^{\frac{3}{2}} = 8 \implies \boxed{x = 3}$$

(c)

$$s = \frac{2}{3}(x+1)^{\frac{3}{2}} - \frac{2}{3}$$
$$s' = \sqrt{x+1} = \sqrt{3+1} = \boxed{2}$$

It is moving 4 units per second along the curve.

Problem 9

(a)

$$y = \frac{x^3}{6} + \frac{1}{2x} \implies y' = \frac{x^2}{2} - \frac{1}{2x^2} \geq 0$$
$$A = \int_1^2 \frac{x^3}{6} + \frac{1}{2x}\,dx$$

$$= \frac{x^4}{24} + \frac{1}{2}\ln x\Big|_1^2$$
$$= \boxed{\frac{5}{8} + \frac{1}{2}\ln 2}$$

(b)

$$l = \int_1^2 \sqrt{1 + \left(\frac{x^2}{2} - \frac{1}{2x^2}\right)^2}\, dx$$
$$= \int_1^2 \sqrt{\left(\frac{x^2}{2} + \frac{1}{2x^2}\right)^2}\, dx$$
$$= \int_1^2 \frac{x^2}{2} + \frac{1}{2x^2}\, dx$$
$$= \frac{x^3}{6} - \frac{1}{2x}\Big|_1^2$$
$$= \boxed{\frac{17}{12}}$$

(c)

$$V = \pi \int_1^2 \left(\frac{x^3}{6} + \frac{1}{2x}\right)^2 dx$$
$$= \pi \int_1^2 \frac{x^6}{36} + \frac{x^2}{6} + \frac{1}{4x^2}\, dx$$
$$= \pi(\frac{x^7}{252} + \frac{x^3}{18} - \frac{1}{4x})\Big|_1^2$$
$$= \boxed{\frac{57\pi}{56}}$$

(d)

$$V = 2\pi \int_1^2 x\left(\frac{x^3}{6} + \frac{1}{2x}\right) dx$$
$$= 2\pi \int_1^2 \frac{x^4}{6} + \frac{1}{2}\, dx$$
$$= 2\pi(\frac{x^5}{30} + \frac{1}{2}x)\Big|_1^2$$
$$= \boxed{\frac{46\pi}{15}}$$

(e) The former represents the straight line distance, which is the strict lower bound, while the latter represent the upper bound, simply the addition of Δx and Δy.

Problem 10

(a)

$$L = 2 \cdot n \cdot \frac{1}{n} = \boxed{2}$$

(b)

$$\begin{aligned} l &= \int_0^{\pi} \sqrt{1 + n^2 \cos^2(n^2 x)}\, dx \\ &\geq \int_0^{\pi} n|\cos(n^2 x)|\, dx = 2n \end{aligned}$$

(c)

$$\begin{aligned} 0 &\leq f'^2 - 2f' + 1 \\ 1 + f'^2 + 2f' &\leq 2f'^2 + 2 \\ 1 + f' &\leq \sqrt{2}\sqrt{1 + (f')^2} \end{aligned}$$

It is exactly $\sqrt{2}$ when $f' = 1$.

(d)

$$\begin{aligned} l &= \int_a^b \sqrt{1 + f'(x)^2}\, dx \\ g(x) = kf\left(\frac{x}{k}\right) &\implies g'(x) = f'\left(\frac{x}{k}\right) \\ l &= \int_{ka}^{kb} \sqrt{1 + f'\left(\frac{x}{k}\right)}\, dx \\ &= k \int_a^b \sqrt{1 + f'(u)^2}\, du \end{aligned}$$